MILITARY SPECIFICATION

CHARGE, PROPELLING, PA500, FOR MINE M16

This specification is approved for use by Picatinny Arsenal, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for the Propelling Charge Assembly for the M16A2 Antipersonnel Mine.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL PPP-B-26 - Bag, Plastic, Polyethylene

MILITARY

MIL-P-116	- Preservation - Packaging, Methods of
MIL-P-223	- Powder, Black
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-B-81705	- Barrier Material Flexible, Electrostatic-Free, Heat Sealable

STANDARDS

FEDERAL

FED-STD-101 -	Preservation, Packaging and Packing Materials Test procedures

FSC: 1345

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MIL-STD- 105 - Sampling Procedures and Tables for Inspection by Attributes (ABC- ST D-105)
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

PICATINNY ARSENAL

76-4-59 - Box, packing Metal-Wood, M18 for smokeless Powder, Assembly
8796343 - Propelling Charge Assembly
7548645 - Box, Fiberboard, Packing, Reusable, Collapsible, For High Explosives
8858848 - Marking Diagram and Sealing of Metal lined Wooden Packing boxes for shipment of Propellants
9241693 - packing and Marking of Fiberboard cartons for shipment of propellant increments

[Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Material. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Propelling Charge Assembly. - The propelling charge assemblies shall comply with all the requirements specified on drawing 8796345 and with all requirements specified in applicable Specifications.

3.3 Moisture Content of Black Powder. - The moisture content of the black powder at the time of loading shall not exceed the percentage specified in Specification MIL-P-223.

3.4 Functioning. - The propelling charge shall propell a fifty (50) pound weight to a minimum height of six and one half (6.5) feet (see 4.5.4).

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3.5 Workmanship. - All parts shall be fabricated and finished or a ricated and loaded) in a thorough, workmanlike manner, and all manufacturing, processing and assembly operations shall be correctly performed. The parts shall be clean and free of burrs, sharp edges, unblended radii, surface defects, chips, dirt, grease and oil (except where specifically required), corrosion products and other foreign matter. The cleaning methods used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.6 First Article Inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. - The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. - The contractor shall submit a first article sample [see 6.1b) as designated by the Contracting Officer. The first article sample shall consist of 50 complete repelling charge assemblies. The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.5.3.

4.3.2 Inspections to be Performed. - See MIL-A-48078 and paragraphs 4,4.2.1 and 4 .5.4 specified herein.

4.3.3 Rejection. - See MIL-A-48078.

4.4 Quality Conformance Inspection

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4.4.1 Inspection Lot Formation. - Inspection lots shall comply with the lot formation provisions to MIL-A-48078. Lot-numbering, as required, shall be in accordance with MIL-STD-1168. In addition, inspection lots shall contain:

a. Black powder of one interfix lot number from one manufacturer.

4.4.2 Examination. - (See MIL-A-48078). - Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48887 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Propelling Charge Assembly				8796343
					NEXT HIGHER ASSEMBLY
					8864746
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of Black Powder		See 4.5.2	3.2	4.5.2
102	Color improper		0.40%	3.2	Visual
103	Bag damaged (hole, tear, weak spots or breaks)		0.40%	3.2	Visual
104	Evidence of poor workmanship		0.40%	3.2	Visual
Minor					
201	Seam stitching inadequate (incomplete, opening, etc)		0.65%	3.2	Visual

NOTES

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48887(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Box, Fiberboard, Packing, Reuseable				7548645
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Liner damaged so that contents are exposed		0.40%	3.2	Visual
102	Bag incompletely sealed		0.40%	3.2	Visual
Minor	None defined				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48887(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Packing and Marking of Fiberboard Cartons for Shipment of Propellant Increments				9241693
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Banding or strapping missing, broken or loose		0.40%	3.2	Visual
102	Board broken or split		0.40%	3.2	Visual
103	Contents exposed		0.40%	3.2	Visual
Minor					
201	Banding or strapping improperly engaged		0.65%	3.2	Visual
202	Contents move when shaken		0.65%	3.2	Visual
203	Markings missing, misleading or illegible		0.65%	3.2	Visual
204	Nails, screws or fasteners loose		0.65%	3.2	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48887 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Box, Packing, Metal-Wood (Prior to filling)				76-4-59
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE / INSPECTION METHOD
Critical	None defined				
Major					
101	Cracks or holes in metal liner		0.40%	3.2	Visual
102	Foreign matter on interior		0.40%	3.2	Visual
Minor	None defined				

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48887(PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Box, Packing, Metal-Wood (Sealed)		76-4-59
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cover incompletely closed		0.40%	3.2	Visual/Manual
102	Gasket missing or broken		0.40%	3.2	Visual
103	Wooden cover split or otherwise damaged		0.40%	3.2	Visual
Minor					
201	Car seal missing, unsealed or improperly positioned		0.65%	3.2	Visual/Manual
202	Crack or hole in metal container		0.65%	3.2	Visual
203	Hardware improperly engaged		0.65%	3.2	Visual
204	Marking misleading or unidentifiable		0.65%	3.2	Visual
205	Bare areas in exterior coating or metal container the sum of which is in excess of 1/4 square inch		0.65%	3.2	Visual

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4. 4.3 Testing

4.4. 3.1 Moisture Content of Black Powder. - (See 3.3) Major Defect, The moisture content of the black powder shall be determined at the time of loading not less than once for each eight hours of production, or portion therof. two grams of black powder shall be selected just prior to loading and the moisture content determined as specified in 4.5.1. Composite samples shall not be" used in this determination. If the moisture content exceeds the requirement specified, the black powder shall be rejected. If this determination is made after loading has begun, components loaded therewith shall be rejected.

4.4.3 Funtioning. - (See 3.4) Major Defect - Fifty propelling charges s al be tested as specified in paragraph 4.5.4 using equipment in accordance with 4.4.4, If one or more assemblies fail to function the lot shall be rejected.

4.4.4 Inspection Equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test Methods and Procedure

4.5.1 Moisture Content of Black Powder. - The moisture content of the black powder shall be determined as specified in Specification MIL-P-223.

4.5.2 Propellant Weight. - (See 5.2) - The inspections below are to be conducted by inspection personnel using equipment designated for inspection purposes and shall be clearly distinguished from production operations. Any increment which fails to comply with the applicable requirement shall be classed defective and removed from the lot. Inspection equipment shall be in accordance with 4.4.4.

4.5.2.1 Method 1. - The loaded propellant charge assemblies will be inspected 100 percent. The weight of the propellant will be determine by subtracting an average weight of the empty bag or the actual weight as marked on the individual bags from the load-

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ed propellant charge assembly weight.

4. 5.2.2 Method 2. - The weight of the propellant will be inspected 100 percent prior to loading the propellant charge assembly on a different balance. In addition, the propellant weight of the loaded assembly will be inspected using a sample plan from MIL-STD-105 or MIL-STD-1235 with an AQL of 0.40 percent. The weight of the propellant will be determined by subtracting an average weight of the empty bag or the actual weight as marked on the individual bag from the loaded weight.

4.5.3 Heat Sealed Seam Test. - This test shall be performed specified in MIL-P-116.

4.5.4 Functioning - The propelling charge assembly shall be funtioned using the flash igniter and primer assembly of the M605 fuze (see dwg. 7548286). The fifty (50) pound weight, the propelling charge and the flash igniter and primer assembly shall be positioned in a fixture which simulates the M16A2 mine configuration (see dwg. 8876013. The cavity required to accept the propelling charge and the flash igniter and primer assembly shall have the same volume as is required for these assemblies in the M16A2 mine. The test fixture, the measuring equipment and the test procedures shall be in accordance with 4.4.4.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A. - Pack approximately 3000 charges in container, Propellant M18. The container shall be lined with two (2) conductive Polyolefin Bags, one within the other and individually tied after filling. The bags shall comply with the requirements of 5.1. 1.1. The container shall be securely closed. The air pressure test is not required. The Gross weight shall not exceed 60 pounds.

5.1.1.1 Bags. - Bags shall be made of conductive Polyethylene or polyolefin sheet with a nominal thickness of .003 inches (tolerence = plus Or minus 20 percent). This surface resistivity shall not exceed 1000 ohms per centimeter, when tested in accordance with test method 4046 of FED-STD-101. The electric bleed-off shall not exceed 2.00 seconds when tested in accordance with test method 4046 of FED-STD-101. Tolerance on lengths and widths shall be as specified herein and shall comply with PPP-B-26. Details not specified herein shall comply with PPP-B-26.

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5. 1.2 Level B. - Propellant charge shall be packed as specified in 5.1.1 Alt erantively, the charges may be packed in the M1 8 container provided the filled container meets the air pressure test called for on dwg. C8858848. The gross weight shall not exceed 60 pounds.

5.1.3 Level C. - Propellant charges may be packed as specified in 5.1.1 or 5.1.2. Alternatively, the charges shall be packed in heat sealed barrier bags made in accordance with the construction requirements for Type I, Class E of specification MIL-B-117 except that the barrier material shall comply with Type I of specification MIL-B-81705. One or two bags, with a total net weight not exceeding 50 pounds shall be overpacked in fiberboard box complying with No. 1 or No. 4 of dwg. C-8796745. Size of barrier material bag shall be 33 open face X 29 deep for size #4 or 30 open face X 29 deep for size #1 box. After loading bags, exhaust all excess air within bag prior to sealing. Seal bag in accordance with manufacturers instructions. The seals shall comply with the heat seal test in accordance with Spec. MIL-P-116. Insert sealed bag into Fiberboard Box, close and seal as shown on drawing C - 7548645.

5.2.1 Propellant Container M18. - Mark in accordance with drawing C 8858848,

5.2.2 Fiberboard Box. - Mark box in accordance with drawing 9241693.

5.2.3 Dot Marking. - All loaded shipping container shall be marked. "Black Powder".

6. NOTES

6.1 Ordering Data. - Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.2 Submission of Inspection Equipment Designs for Approval. - (See MIL-A-48078). Submit equipment designs as required to Commander, Picatinny Arsenal, ATTN: SAEPA-QA-T-T, Dover, New Jersey 07801.

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6.3 Distribution of Ammunition Data Cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801.

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Preparing Activity
Army-PA

Project Number: 1345-A-156

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